ACCESS CONTROL LIST (ACL)

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Project Name	**Script-Controlled ACL – Restrict Record Access Based on Field Value**
Maximum Marks	3 Marks

ACCESS CONTROL LIST (ACL) MATRIX

Script-Controlled ACL – Restrict Record Access Based on Field Value

The ACL matrix defines the access permissions configured for the Institution Details table in the ServiceNow Developer Instance. Four separate ACLs are created for Read, Create, Write, and Delete operations. Each operation is associated with a specific role to provide controlled access to records.

1. ACL Permission Matrix

ACL No.	Operation	Table	Required Role	Condition / Control	Purpose
ACL-01	Read	Institution Details	BB1	Branch = EEE + script-controlled authorization	Controls which records the authorized user can view.
ACL-02	Create	Institution Details	BB2	Role-based authorization	Controls creation of new Institution Details records.
ACL-03	Write	Institution Details	BB3	Role-based authorization	Controls modification of existing records.
ACL-04	Delete	Institution Details	BB4	Role-based authorization	Controls deletion of existing records.

2. Role-to-Operation Mapping

Role	ACL Operation	Access Function
BB1	Read	View authorized Institution Details records
BB2	Create	Create new Institution Details records
BB3	Write	Modify Institution Details records
BB4	Delete	Delete Institution Details records

3. Field-Based Read Restriction

The Read ACL is the main field-based security mechanism in the project. The ACL is configured for the Institution Details table with role BB1 and a Branch condition. The project demonstrates the condition Branch = EEE. A script is used in the ACL authorization section to provide additional authorization based on the Branch field value.

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ACL MATRIX – ACCESS VALIDATION AND EXPECTED RESULTS

4. Institution Details Data Used for Testing

Student ID	Student	Branch	Access Relevance
100	Student A	ECE	Restricted for EEE-only read access
200	Student B	EEE	Authorized / visible to configured EEE user
300	Student C	CSC	Restricted for EEE-only read access

5. Access Validation Matrix

User / Role	Requested Operation	Branch / Record	Expected Result
Project User (BB1)	Read	EEE	Access permitted
Project User (BB1)	Read	ECE	Access restricted
Project User (BB1)	Read	CSC	Access restricted
Authorized User (BB2)	Create	Institution Details	Create permitted when role requirement is satisfied
Authorized User (BB3)	Write	Institution Details	Modification permitted when role requirement is satisfied
Authorized User (BB4)	Delete	Institution Details	Deletion permitted when role requirement is satisfied
Administrator	Administrative access	All Institution Details records	Full administrative access for management and verification

6. Security Control Flow

1. The user requests access to an Institution Details record or list.
2. ServiceNow identifies the requested operation: Read, Create, Write, or Delete.
3. The corresponding ACL checks the required role.
4. For the Read operation, the configured Branch condition and script-controlled authorization are evaluated.
5. If the user and record satisfy the configured security rules, access is permitted; otherwise, access is restricted.
6. The project validates the result by checking the Institution Details list using the configured project user.

7. Expected Security Outcome

The completed ACL configuration provides operation-specific access control for the Institution Details table. The project demonstration confirms the field-based Read restriction by showing the EEE branch record to the configured user while records from other branches are restricted. The separate Create, Write, and Delete ACLs provide additional role-based controls. This design supports controlled record access and the principle of least privilege within the ServiceNow application.